

WESTERN REFRIGERATION PVT. LTD.

System Architecture

Product: WRL Portal

Version: 1.0 draft

Date: 2026-09-02

Production: <https://wrl-dashboard.vercel.app>

Internal — Handover Documentation

Status: Ready (repo-generated). For printable diagrams see `07-Company-Share/PDF/03_Architecture_Diagrams.pdf` .

WRL Portal — Architecture Documentation

Product: WRL Portal (WRL Dashboard)
Stack: Next.js · Supabase (auth/profiles) · Postgres read-model (VPS) · CRM (trhcalls)
Repo: `fast-close-app`

0. Start Here

Run locally

```
npm install
npm run dev           # Next.js on localhost:3000
# Optional – point READ_SUMMARY_FROM=postgres in .env.local to query the VPS Postgres instead of C
```

For VPS workers (sync + email) see `scripts/vps-hosting/README.md` .

Reading order for a new developer

Step	Read	Why
1	§1 Module Map	Understand which modules exist and how they connect
2	§6 Deployment / Infrastructure	Know where each piece actually runs
3	§7 ETL Data Flow	Understand CRM → hot table → portal → output
4	§8 Background Jobs	Know what runs on the VPS and when
5	§2.1–2.2 Auth + MIS Load	Two most common user flows

Step	Read	Why
6	§2.8–2.12 (cancelled, Athena, digests, subcontractor, attendance)	Newer report + ops flows
7	§10 RBAC Flow	Before touching permissions or API routes
8	§11 Failure Paths	Before touching production

Diagram assets: docs/diagrams/ — 20 Mermaid sources (`.mmd`) + PNG renders. Regenerate with `node scripts/ops/export-mermaid-diagrams.mjs` after editing diagrams in this file.

0.1 Known Limitations & Design Decisions

These are deliberate simplifications or known gaps — not bugs to fix immediately.

Area	Decision / Limitation	Ceiling / Upgrade path
RLS	Supabase Row-Level Security is not used; access is enforced in application code via RBAC catalog + office-scope filter on every query	Enable RLS policies when multi-tenant isolation becomes a hard requirement
Read-model staleness	<code>calls_latest_hot</code> can lag CRM by up to ~3 min (sync cycle); call detail drawer always hits CRM live	Acceptable for reporting; not suitable for real-time ops
No <code>.test.tsx</code> coverage	UI components have near-zero browser test coverage; server logic has targeted unit tests only (<code>mail-basis</code> , <code>major-repair-alert</code>)	Add Playwright or Vitest browser tests when UI stabilises
Pagination inconsistency	Some API routes return all rows (register export), others paginate. No unified cursor scheme	Standardise to cursor-based pagination if row counts grow past ~50k
One-PR-per-slice	Module boundary discipline is aspirational; some shared logic still lives in root <code>src/lib</code> rather than being owned by one module	Enforce via lint rule (<code>no-cross-module-import</code>) when team grows

Area	Decision / Limitation	Ceiling / Upgrade path
Email transport	All outbound mail (MIS digest, cancelled digest, major-repair, subcontractor) uses <code>sendDigestPayload()</code> / <code>sendHtmlEmail()</code> in <code>mis-email/services/send.ts</code> — relay on Vercel, direct SMTP on VPS	
Email retry	Failed digest sends are logged but not retried automatically; next cron poll (15 min) is the retry	Add a send-retry queue if delivery SLA tightens
CRM is MS SQL / read-only	All CRM writes go through the CRM application, not the portal; portal is strictly a reporting read layer	
VPS Postgres is not HA	Single-node Postgres on Hostinger VPS; no replica or auto-failover	Migrate to managed Postgres (Supabase, RDS) if uptime SLA requires 99.9%+

Rule: Update this document only when the codebase changes (schema migration, new module, new cron job, RBAC change). Do not update speculatively.

1. System Workflow — Module Map

Diagram: See *Architecture Diagrams* PDF (`03_Architecture_Diagrams.pdf`) or `07-Company-Share/Diagrams/` .

2. Sequence Diagrams

2.1 — User Authentication Flow

Diagram: See *Architecture Diagrams* PDF (`03_Architecture_Diagrams.pdf`) or `07-Company-Share/Diagrams/` .

2.2 — MIS Report Page Load

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

2.3 — MIS Email Digest (Cron / Automated)

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

2.4 — Manual MIS Email Compose & Send (Profile Page)

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

2.5 — Major Repair Repeat Alert

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

2.6 — CRM → Postgres Read-Model Sync Pipeline

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

2.7 — ARCP Claims Load (Hybrid Postgres ↔ CRM)

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-

Share/Diagrams/ .

2.8 — Cancelled Calls Report

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

2.9 — Athena Failed-Calls Reconciliation

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

2.10 — Cancelled Call Digest (Evening Ops)

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

2.11 — Subcontractor Stock (SAP vs CRM)

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

2.12 — Service Call Activity (Admin)

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

3. Data Layer Reference

Table / Source	Owner	Purpose
<code>calls_latest_hot</code>	<code>@/lib/read-model</code>	CRM call mirror, incremental sync
<code>calls_cancelled</code>	<code>@/lib/read-model/cancelled-call-register</code>	Cancelled call register (Postgres)
<code>athena_failed_calls_raw</code> / <code>_normalized</code>	<code>@/lib/read-model/athena-reconciliation</code>	Athena API failure ingest + match status
<code>crm_attendance_details</code>	<code>@/lib/read-model/attendance-details</code>	Technician activity mirror for admin report
<code>mis_client_import_rows</code>	<code>mis</code> client-import	Uploaded client MIS batches
<code>arcp_lines_hot</code>	<code>arcp-claims/sync</code>	ARCP claim lines hot cache
<code>cancelled_call_digest_recipients</code>	<code>mis-email/sync</code>	Per-branch cancelled digest To list
<code>subcontractor_stock_*</code> / <code>subcontractor_sap_mail_log</code>	<code>subcontractor-stock</code>	SAP reconcile config, runs, inbox log
<code>app_users</code>	Supabase + admin	Portal users, roles, preferences
<code>call_comments</code>	<code>calls</code> module	Portal-only triage notes
<code>call_flags</code>	<code>calls</code> module	Portal triage flags (noted/escalate/query)
<code>mis_email_routing_rules</code>	<code>mis-email</code>	Routing rule schedule + To/Cc
<code>mis_email_routing_send_log</code>	<code>mis-email</code>	Slot dedupe log
<code>major_repair_recipients</code>	<code>mis-email/sync</code>	Alert recipient rows per branch
<code>security_audit_log</code>	<code>@/lib/security</code>	All sensitive actions
CRM <code>trhcalls</code>	External CRM	Source of truth for all calls

4. Permission Gates Summary

Feature	Required Permission
MIS Summary tab	tab_mis_summary
MIS Register tab	tab_mis_register
MIS Accounts tab	tab_mis_accounts
ARCP Claims page	page_arcp_claims
Distribution page	page_distribution
Serial Audit	page_serial_audit
Location Audit	page_location_audit
Warranty Master	page_warranty_master
Athena failed-calls	page_athena_reconciliation
Cancelled Calls	page_cancelled_calls (also granted by page_athena_reconciliation)
MIS Email send	mis_email_reports
Mail & Alerts (routing, digests, subcontractor)	page_mis_email_settings
Service call activity	manage_users (hard-coded /admin/attendance)
Admin users	manage_users
VPS Cron manage	super_admin
Performance insights	page_performance_insights
HOD (all offices)	view_all_offices

5. Module Quick-Reference

Per-module design notes: `src/modules/<name>/README.md` . MIS sub-leaves: `mis/register/` , `mis/client-import/` .

Module	Path	Key responsibility
<code>auth</code>	<code>src/modules/auth</code>	Sign-in/out, /me, password reset
<code>mis</code>	<code>src/modules/mis</code>	Reports: summary, register, BD-MIS, client import
<code>calls</code>	<code>src/modules/calls</code>	Call detail drawer (CRM), comments, flags, offices
<code>mis-email</code>	<code>src/modules/mis-email</code>	Digest scheduler, compose, SMTP send, major-repair + cancelled-call alerts
<code>arcp-claims</code>	<code>src/modules/arcp-claims</code>	ARCP register + hybrid hot/CRM load
<code>sync</code>	<code>src/modules/sync</code>	Read-model status API, VPS cron pause API
<code>performance</code>	<code>src/modules/performance</code>	Web Vitals + server snapshot
<code>distribution</code>	<code>src/modules/distribution</code>	Call distribution + idle assignees
<code>serial-audit</code>	<code>src/modules/serial-audit</code>	Repeat complaint serial history
<code>location-audit</code>	<code>src/modules/location-audit</code>	Tech GPS vs install address
<code>warranty-master</code>	<code>src/modules/warranty-master</code>	Warranty coverage reports
<code>cancelled-calls</code>	<code>src/modules/cancelled-calls</code>	Cancelled register (Postgres), CSV/Excel export, digest data
<code>athena-reconciliation</code>	<code>src/modules/athena-reconciliation</code>	Athena API failure triage + reconcile UI
<code>attendance</code>	<code>src/modules/attendance</code>	Service call activity admin report
<code>subcontractor-stock</code>	<code>src/modules/subcontractor-stock</code>	SAP vs CRM stock reconcile + email (VPS)
<code>roles</code>	<code>src/modules/roles</code>	Role + permission admin UI
<code>users</code>	<code>src/modules/users</code>	User admin UI

Module	Path	Key responsibility
security-audit	src/modules/security-audit	Audit log viewer

6. Deployment / Infrastructure Diagram

Where each piece physically runs and how they talk to each other.

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

Key facts:

- Next.js runs on **Vercel** (serverless); no long-running processes.
- All background workers and crons run on the **Hostinger VPS**.
- Email from Vercel goes over HTTPS to a VPS relay server, which forwards via Postfix (avoids Vercel's SMTP restriction).
- **All outbound mail** (MIS, cancelled digest, major-repair, subcontractor) shares `sendDigestPayload()` / `sendHtmlEmail()` in `mis-email/services/send.ts` .
- Unified `*/15` mail cron: `mail-scheduler.sh` runs MIS digest + subcontractor steps (see [MAIL_SCHEDULE.md](#)).
- Supabase handles **auth only** (GoTrue + app_users); the read-model Postgres is a separate VPS database.

7. Overall Data Flow / ETL

Single picture: CRM → Postgres hot tables → Portal queries → Excel / Email output.

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

8. Background Jobs / Cron Map

Every scheduled or daemon job, its schedule, pause flag, and log location.

Mail-specific schedule: [MAIL_SCHEDULE.md](#) (canonical cron table).

Sync CLI map: [SYNC_ENTRY_POINTS.md](#) .

Diagram: See *Architecture Diagrams* PDF ([03_Architecture_Diagrams.pdf](#)) Or [07-Company-Share/Diagrams/](#) .

Job	Schedule (IST)	Days	Pause Flag	Log
CRM incremental sync	Every ~3 min (systemd daemon)	Mon–Sun	<code>crm_sync</code> (portal VPS Cron)	<code>logs/sync-worker.log</code>
Nightly sidecar sync	02:30 (systemd timer)	Mon–Sun	—	<code>logs/sync-worker-nightly.log</code>
Mail scheduler (<code>mail-scheduler.sh</code>)	Every 15 min poll (cron)	Mon–Sun (MIS skips Sunday IST inside script)	<code>mis_email_digest</code> + <code>subcontractor_stock</code> per step	<code>logs/mail-scheduler.log</code>
MIS email test digest	Ad-hoc cron (optional)	Mon–Sat	<code>mis_email_test</code> (portal VPS Cron)	<code>logs/mis-email-cron.log</code>
Morning watchdog	09:50	Mon–Sat	—	<code>logs/mis-email-watchdog.log</code>
Subcontractor stock	Via mail-scheduler step 2	Mon–Sun	<code>subcontractor_stock</code> (portal VPS Cron)	<code>logs/mail-scheduler.log</code>
Evening ops probe	16:00	Mon–Sun	Per-step in TS (<code>mis_email_test</code> , <code>cancelled_call_digest</code> , <code>subcontractor_stock</code>)	<code>logs/evening-ops-sequencer.log</code>
Cancelled-call digest	Via evening-ops (not standalone */15)	Mon–Sat	<code>cancelled_call_digest</code>	<code>logs/evening-ops-sequencer.log</code>

Job	Schedule (IST)	Days	Pause Flag	Log
Client import file purge	03:15 daily	Mon–Sun	—	logs/mis-client-purge.log
VACUUM FULL	00:00 Sunday	Sun	—	logs/vacuum-full.log
ARCP nightly sync	Via arcp-nightly.ps1 / CLI	Nightly	—	stdout

Diagram: See *Architecture Diagrams* PDF (03_Architecture_Diagrams.pdf) OR 07-Company-Share/Diagrams/ .

Portal pause gates (bash vs TypeScript)

Job / step	Pause ID	Bash gate (vps-cron-gate.sh)	TS gate (isVpsCronPaused)
MIS digest cron	mis_email_digest	Yes — mis-email-digest.sh	Inside runMisEmailDigest
Subcontractor cron	subcontractor_stock	Yes — subcontractor-stock-cron.sh	Inside subcontractor CLI
Mail scheduler	(per step above)	Delegates to child scripts	Delegates to child scripts
Evening ops — MIS test	mis_email_test	No	Yes — evening-ops-sequencer.ts
Evening ops — cancelled digest	cancelled_call_digest	No	Yes
Evening ops — subcontractor	subcontractor_stock	No	Yes
CRM sync worker	crm_sync	No (systemd)	N/A

9. Key Tables — Simplified ERD

Diagram: See *Architecture Diagrams* PDF (`03_Architecture_Diagrams.pdf`) Or `07-Company-Share/Diagrams/` .

10. RBAC Permission Decision Flow

How every API request resolves access — from cookie to allow/deny.

Diagram: See *Architecture Diagrams* PDF (`03_Architecture_Diagrams.pdf`) Or `07-Company-Share/Diagrams/` .

11. Failure & Degradation Paths

What actually happens when each external dependency goes wrong.

Diagram: See *Architecture Diagrams* PDF (`03_Architecture_Diagrams.pdf`) Or `07-Company-Share/Diagrams/` .

Failure	Impact	Recovery
CRM down	Dashboard serves stale hot data; call drawer fails	Retry next sync cycle; no data loss
Sync lock stuck	Hot table drifts until lock released	<code>systemctl restart fast-close-sync-worker</code> or cancel Postgres backend
SMTP fail	Digest not sent; watchdog fires	Re-run via <code>npm run mis-email:preview</code> or next 15-min cron
VPS Cron paused	All cron jobs skip silently	Resume via <code>/admin/vps-cron</code> (super_admin)
Postgres slow	Dashboard timeouts; slow exports	Run VACUUM ANALYZE; check seq scan indexes

Failure	Impact	Recovery
Vercel deploy fails	UI unreachable	Instant rollback; VPS workers unaffected
Supabase auth down	Login blocked; cookie sessions degrade	Cookie fallback for existing sessions
SAP file missing	Subcontractor emails skipped for the day	Wait for inbox sync; check <code>/tmp/extracted_sap</code> on VPS
Athena/cancelled sidecar lag	Reports show stale data	Re-run nightly timer or trigger sync from UI